

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Product Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Owned product-facing automation features in a small team, turning ambiguous MCA filing steps into reliable Node.js/Puppeteer workflows with operator checkpoints.
- Built backend validation, retries, public-site API lookups, and recoverable session state, reducing manual filing time by **85%**.
- Shipped Dockerized services on Google Cloud Run with GitHub Actions releases, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Contributed production-grade Python features inside a mature open-source application, coordinating model outputs, UI paths, generated artifacts, and user inspection workflows.
- Reduced expensive image-processing runtime by **87%** through async/parallel execution while preserving correctness on large data paths.
- Worked through unfamiliar code quickly across VTK, wxPython, NumPy, PyTorch/ONNX-adjacent inference, and label-mapping logic.

Open Source

Contributions 

- Contributed maintainer-reviewed patches across **Invesalio**, **IOOS**, **VideoLAN**, **CRIU**, **Tiled**, **Kyverno**, and **Kuadrant MCP Gateway**, including medical-imaging workflow fixes, parser robustness, Zip Slip path validation, and cloud-native reliability paths.

Projects

Seaweed: Contest Platform | *TypeScript, React, Next.js, Go, PostgreSQL, Judge0, Firebase, Docker*

GitHub 

- Built an end-to-end assessment product with auth-gated problem pages, code editor, submissions, admin review, shortlisting, live rankings, and reproducible judging records.
- Implemented Go judging APIs with Judge0, PostgreSQL rankings/materialized views, Redis-backed runtime state, and Docker/Kubernetes deployment for **500+ concurrent users**.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

GitHub 

- Built browser-automation-backed product workflows with REST APIs, RabbitMQ workers, Redis autosave, PostgreSQL persistence, durable job status, and recoverable publishing tasks.
- Added retries, DLQs, circuit-breaker isolation, health checks, and Prometheus metrics so long-running API/browser workflows stayed observable and fault-tolerant.

Hyoka | *Python, FastAPI, PostgreSQL, Next.js, TypeScript, OpenTelemetry, Docker*

- Built backend and dashboard paths for operating agent behavior, including trace ingestion, suites, candidates, validation runs, promotion state, audit history, and gate decisions.
- Designed project-scoped API keys, worker leases, signed manifests, JSON schema checks, replay artifacts, and containerized Postgres-backed services.

Technical Skills

Languages: TypeScript, JavaScript, Python, Go, SQL

Frameworks: React, Next.js, Node.js, FastAPI, Echo, PyTorch

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, GCP, AWS, GitHub Actions, OpenTelemetry, Prometheus